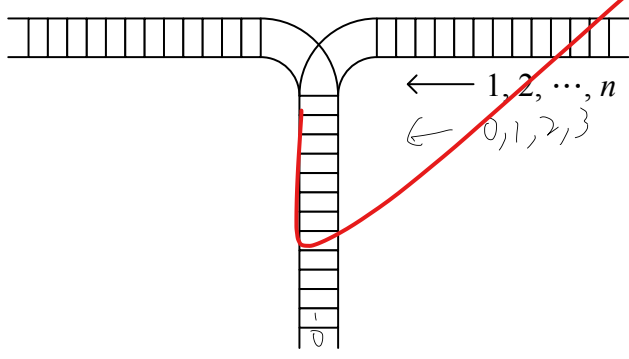


Homework 2.

100

- The file name of your homework (in PDF) should be in the format: “學號-作業編號.pdf”. For example: 00357999-hw2.pdf
- Please submit your homework to Tronclass before **2024/10/27 23:59**.
- **So, we do NOT accept late submission for this homework.**

1. (10%) Consider the railroad switching network. Railroad cars numbered $0, 1, 2, \dots, n-1$ are at the right. Each car is brought into the stack and removed at any time. For instance, if $n = 3$, we could move in 0, move in 1, move in 2, and then take the cars out, producing the new order 2, 1, 0. Find all possible permutations for $n = 4$.



- ④
- 0123
 - 0132
 - 0213
 - 0231
 - 0321
 - 1023
 - 1032
 - 1203
 - 1230
 - 1320
 - 2103
 - 2130
 - 2310
 - 3210

2. (20%) Given a **stack** of size 4, according to Program 3.1 to 3.3 in the textbook, what is the stack in each step when we do the following commands:

push J1, push J2, push J3, pop, push J4, push J5, push J6

top	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1					stack is empty
0	J1				J1 is added
1	J1	J2			J2 is added
2	J1	J2	J3		J3 is added
1	J1	J2			J3 is deleted
2	J1	J2	J4		J4 is added
3	J1	J2	J4	J5	J5 is added
3	J1	J2	J4	J5	stack is full

3. (20%) Given a **queue** of size 4, according to Program 3.5 to 3.6 in the textbook, what is the stack in each step when we do the following commands:

add J1, add J2, add J3, delete, add J4, add J5, add J6

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1	-1					queue is empty
-1	0	J1				J1 is added
-1	1	J1	J2			J2 is added
-1	2	J1	J2	J3		J3 is added
0	2		J2	J3		J1 is deleted
0	3		J2	J3	J4	J4 is added
0	3		J2	J3	J4	queue is full
0	3		J2	J4	J4	queue is full

4. (20%) Given a **circular queue** of size 4, according to Program 3.7 to 3.8 in the textbook, what is the stack in each step when we do the following commands:

add J1, add J2, add J3, delete, add J4, add J5, delete, delete

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
0	0					queue is empty
0	1		J1			J1 is added
0	2		J1	J2		J2 is added
0	3		J1	J2	J3	J3 is added
1	3			J2	J3	J1 is deleted
1	0	J4		J2	J3	J4 is added
1	0	J4		J2	J3	queue is full
2	0	J4			J3	J2 is deleted
3	0	J4				J3 is deleted

5. (20%) Write the postfix form of the following infix expressions:

- (a) $(A + B) * C + D / (E + F * G) - H$ (a) $AB+C \times DEFG \times + / + H -$
 (b) $(A \&\& B) \parallel C \parallel (E > F)$ (b) $AB \&\& C \parallel EF > ! \parallel$

6. (10%) Write the infix form of the following postfix expression:

$ABC < CD > \parallel ! \&\& ! CE < \parallel$

$!(A \&\& ((B < C) \parallel (C > D))) \parallel (CE)$

